

Principles of Programming Languages

Lecture 7: Small-step Structural Operational Semantics

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Outline

Small-step SOS

- States, state operations, rule schemata (in part 1)

- Difference w.r.t. Big-step SOS (in part 1)

Rules

- Arithmetic expressions (in part 1)

- Boolean expressions (in part 2)

- Statements (in part 2)

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Part 1

Recall: states

- ▶ The **state** is a *partial finite-domain function*:

$$\sigma : Var \rightarrow Int$$

Recall: states

- ▶ The **state** is a *partial finite-domain function*:

$$\sigma : \text{Var} \rightarrow \text{Int}$$

- ▶ Example: $\sigma = x \mapsto 10, y \mapsto 0$

Recall: state operations

- ▶ *Lookup*

$$_(_) : \textit{State} \times \textit{Var} \rightarrow \textit{Int}$$

Recall: state operations

- ▶ *Lookup*

$$_(_) : \textit{State} \times \textit{Var} \rightarrow \textit{Int}$$

- ▶ *Update*

$$_[_/_] : \textit{State} \times \textit{Int} \times \textit{Var} \rightarrow \textit{State}$$

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- ▶ Direct control over what to execute and when

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- ▶ Since big-step defines all computation steps in one transition it cannot capture non-determinism by default
- ▶ Direct control over what to execute and when
- ▶ Therefore, small-step SOS is finer-grain than big-step SOS

Small-step SOS

Ingredients:

- ▶ *Configuration*
 - ▶ Tuple containing semantic ingredients
 - ▶ Examples: $\langle a, \sigma \rangle$, $\langle b, \sigma \rangle$, $\langle s, \sigma \rangle$, $\langle p \rangle$, ...

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- ▶ *Sequent (or transition)*
 - ▶ Pair of configurations, to be *derived* or *proved*
 - ▶ Examples: $\langle a_1, \sigma \rangle \rightarrow \langle a'_1 \rangle$, $\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a'_1 + a_2 \rangle$, ...
 - ▶ Note that we use “ $\rightarrow$ ” instead of “ $\Downarrow$ ” (as for Big-step)

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 - ▶ Examples: $\langle a_1, \sigma \rangle \rightarrow \langle a'_1 \rangle$, $\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a'_1 + a_2 \rangle$, ...
 - ▶ Note that we use “ $\rightarrow$ ” instead of “ $\Downarrow$ ” (as for Big-step)
- ▶ *Rule*
 - ▶ Tells us how to derive sequents
 - ▶ Example:

$$\frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1 \rangle}{\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a'_1 + a_2 \rangle}$$

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Rule schemata

- ▶ Rule schemata:

$$\frac{C_1 \rightarrow C'_1 \quad C_2 \rightarrow C'_2 \quad \cdots \quad C_n \rightarrow C'_n}{C_0 \rightarrow C'_0}$$

Small-step rules for IMP – 1: arithmetic expressions

SMALLSTEP-LOOKUP: $\langle x, \sigma \rangle \rightarrow \langle \sigma(x), \sigma \rangle$ if $\sigma(x) \neq \perp$

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SMALLSTEP-ADD: $\langle i_1 + i_2, \sigma \rangle \rightarrow \langle i_1 + \text{Int } i_2, \sigma \rangle$

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SMALLSTEP-ADD: $\langle i_1 + i_2, \sigma \rangle \rightarrow \langle i_1 + \text{Int } i_2, \sigma \rangle$

Note: + is non-deterministic!

Simple derivation

- ▶ Let us consider $\sigma(x) = 10$. Here is a derivation for sequent $\langle 2 + x, \sigma \rangle \rightarrow \langle 2 + 10, \sigma \rangle$:

$$\frac{\overline{\langle x, \sigma \rangle \rightarrow \langle 10, \sigma \rangle} \text{ SMALLSTEP-LOOKUP}}{\langle 2 + x, \sigma \rangle \rightarrow \langle 2 + 10, \sigma \rangle} \text{ SMALLSTEP-ADD2}$$

- ▶ What about $\langle 2 + x, \sigma \rangle \rightarrow \langle 12, \sigma \rangle$?

Sequences

- ▶ Reflexive and transitive closure $\rightarrow^*$:

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$$\frac{}{a \rightarrow^* a} \text{REFL}$$

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- ▶ Reflexive and transitive closure $\rightarrow^*$:

$$\frac{}{a \rightarrow^* a} \text{REFL}$$

$$\frac{a_1 \rightarrow a_2 \quad a_2 \rightarrow^* a_3}{a_1 \rightarrow^* a_3} \text{TRANS}$$

Derivation example

- We can now prove $\langle 2 + x, \sigma \rangle \rightarrow^* \langle 12, \sigma \rangle$:

$$\frac{\frac{\frac{}{\langle x, \sigma \rangle \rightarrow \langle 10, \sigma \rangle} \text{LOOKUP}}{\langle 2 + x, \sigma \rangle \rightarrow \langle 2 + 10, \sigma \rangle} \text{ADD2} \quad \frac{\frac{\frac{}{\langle 2 + 10, \sigma \rangle \rightarrow \langle 12, \sigma \rangle} \text{ADD}}{\langle 2 + 10, \sigma \rangle \rightarrow^* \langle 12, \sigma \rangle} \text{REFL TRANS}}{\langle 2 + x, \sigma \rangle \rightarrow^* \langle 12, \sigma \rangle} \text{TRANS}$$

Part 2

Small-step rules for IMP – 2: boolean expressions

Less or equal:

$$\text{SMALLSTEP-LEQ1: } \frac{\langle a_1, \sigma \rangle \rightarrow \langle a'_1, \sigma \rangle}{\langle a_1 \leq a_2, \sigma \rangle \rightarrow \langle a'_1 \leq a_2, \sigma \rangle}$$

Small-step rules for IMP – 2: boolean expressions

Less or equal:

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$$\text{SMALLSTEP-LEQ2: } \frac{\langle a_2, \sigma \rangle \rightarrow \langle a'_2, \sigma \rangle}{\langle i_1 \leq a_2, \sigma \rangle \rightarrow \langle i_1 \leq a'_2, \sigma \rangle}$$

Small-step rules for IMP – 2: boolean expressions

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$$\text{SMALLSTEP-LEQ: } \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle i_1 \leq_{Int} i_2, \sigma \rangle$$

Small-step rules for IMP – 2: boolean expressions

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$$\text{SMALLSTEP-LEQ: } \langle i_1 \leq i_2, \sigma \rangle \rightarrow \langle i_1 \leq_{\text{Int}} i_2, \sigma \rangle$$

Note: i_1 is used to ensure sequential strictness.

Small-step rules for IMP – 2: boolean expressions

SMALLSTEP-NOT:

$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

Small-step rules for IMP – 2: boolean expressions

SMALLSTEP-NOT:
$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

SMALLSTEP-NOTTRUE:
$$\langle !\text{true}, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

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SMALLSTEP-AND1:
$$\frac{\langle b_1, \sigma \rangle \rightarrow \langle b'_1, \sigma \rangle}{\langle b_1 \text{ and } b_2, \sigma \rangle \rightarrow \langle b'_1 \text{ and } b_2, \sigma \rangle}$$

Small-step rules for IMP – 2: boolean expressions

$$\text{SMALLSTEP-NOT:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle !b, \sigma \rangle \rightarrow \langle !b', \sigma \rangle}$$

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$$\text{SMALLSTEP-ANDFALSE:} \quad \langle \text{false and } b_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

Small-step rules for IMP – 2: boolean expressions

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$$\text{SMALLSTEP-ANDFALSE:} \quad \langle \text{false and } b_2, \sigma \rangle \rightarrow \langle \text{false}, \sigma \rangle$$

$$\text{SMALLSTEP-ANDTRUE:} \quad o \langle \text{true and } b_2, \sigma \rangle \rightarrow \langle b_2, \sigma \rangle$$

Example

$$\frac{\frac{\frac{}{\langle 1 + 3, \sigma \rangle \rightarrow \langle 4, \sigma \rangle} \text{ADD}}{\langle 1 + 3 \leq 5, \sigma \rangle \rightarrow \langle 4 \leq 5, \sigma \rangle} \text{LEQ}_1 \quad \frac{\frac{\frac{}{\langle 4 \leq 5, \sigma \rangle \rightarrow \langle \text{true}, \sigma \rangle} \text{LEQ}}{\langle \text{true}, \sigma \rangle \rightarrow^* \langle \text{true}, \sigma \rangle} \text{REFL}}{\langle 4 \leq 5, \sigma \rangle \rightarrow^* \langle \text{true}, \sigma \rangle} \text{TRANS}}{\langle 1 + 3 \leq 5, \sigma \rangle \rightarrow^* \langle \text{true}, \sigma \rangle} \text{TRANS}$$

Small-step rules for IMP – 3: statements

$$\text{SMALLSTEP-ASSIGN2:} \quad \frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x = a; , \sigma \rangle \rightarrow \langle x = a'; , \sigma \rangle}$$

Small-step rules for IMP – 3: statements

$$\text{SMALLSTEP-ASSIGN2:} \quad \frac{\langle a, \sigma \rangle \rightarrow \langle a', \sigma \rangle}{\langle x = a; , \sigma \rangle \rightarrow \langle x = a'; , \sigma \rangle}$$

$$\text{SMALLSTEP-ASSIGN:} \quad \langle x = i; , \sigma \rangle \rightarrow \langle \text{skip}, \sigma[i/x] \rangle \quad \text{if } \sigma(x) \neq \perp$$

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$$\text{SMALLSTEP-SEQ1:} \quad \frac{\langle s_1, \sigma \rangle \rightarrow \langle s'_1, \sigma' \rangle}{\langle s_1 \ s_2, \sigma \rangle \rightarrow \langle s'_1 \ s_2, \sigma' \rangle}$$

Small-step rules for IMP – 3: statements

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$$\text{SMALLSTEP-SKIP:} \quad \langle \text{skip } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

Small-step rules for IMP – 3: statements

SMALLSTEP-IF1:

$$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

Small-step rules for IMP – 3: statements

SMALLSTEP-IF1:

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SMALLSTEP-IFTRUE:

$$\langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

Small-step rules for IMP – 3: statements

$$\text{SMALLSTEP-IF1:} \quad \frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$$

$$\text{SMALLSTEP-IFTRUE:} \quad \langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$$

$$\text{SMALLSTEP-IFFALSE:} \quad \langle \text{if } (\text{false}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$$

Small-step rules for IMP – 3: statements

SMALLSTEP-IF1:	$\frac{\langle b, \sigma \rangle \rightarrow \langle b', \sigma \rangle}{\langle \text{if } (b) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle \text{if } (b') \ s_1 \text{ else } s_2, \sigma \rangle}$
SMALLSTEP-IFTRUE:	$\langle \text{if } (\text{true}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_1, \sigma \rangle$
SMALLSTEP-IFFALSE:	$\langle \text{if } (\text{false}) \ s_1 \text{ else } s_2, \sigma \rangle \rightarrow \langle s_2, \sigma \rangle$
SMALLSTEP-WHILE:	$\langle \text{while } (b) \ s, \sigma \rangle \rightarrow \langle \text{if } (b) \ \{s \ \text{while } (b) \ s\} \text{ else skip}, \sigma \rangle$

Derivation example

- Context: $\sigma(y) = 0$
- Derivation with labels (unreadable):

$$\begin{array}{c}
 \frac{}{\langle y, \sigma \rangle \rightarrow \langle 0, \sigma \rangle} \text{LOOKUP} \quad \frac{}{\langle 0 + 10, \sigma \rangle \rightarrow \langle 10, \sigma \rangle} \text{ADD} \quad \frac{}{\langle x = 10, \sigma \rangle \rightarrow \langle \text{skip}, [10/x]\sigma \rangle} \text{ASSIGN} \quad \frac{}{\langle \text{skip}, [10/x]\sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \text{REFL} \\
 \frac{}{\langle y + 10, \sigma \rangle \rightarrow \langle 0 + 10, \sigma \rangle} \text{ADD1} \quad \frac{}{\langle x = 0 + 10, \sigma \rangle \rightarrow \langle x = 10, \sigma \rangle} \text{ASSIGN2} \quad \frac{}{\langle x = 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \text{TRANS} \\
 \frac{}{\langle x = y + 10, \sigma \rangle \rightarrow \langle x = 0 + 10, \sigma \rangle} \text{ASSIGN2} \quad \frac{}{\langle x = 0 + 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \text{TRANS} \\
 \hline
 \langle x = y + 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle
 \end{array}$$

- Derivation without labels:

$$\begin{array}{c}
 \frac{}{\langle y, \sigma \rangle \rightarrow \langle 0, \sigma \rangle} \quad \frac{}{\langle 0 + 10, \sigma \rangle \rightarrow \langle 10, \sigma \rangle} \quad \frac{}{\langle x = 10, \sigma \rangle \rightarrow \langle \text{skip}, [10/x]\sigma \rangle} \quad \frac{}{\langle \text{skip}, [10/x]\sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \\
 \frac{}{\langle y + 10, \sigma \rangle \rightarrow \langle 0 + 10, \sigma \rangle} \quad \frac{}{\langle x = 0 + 10, \sigma \rangle \rightarrow \langle x = 10, \sigma \rangle} \quad \frac{}{\langle x = 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \\
 \frac{}{\langle x = y + 10, \sigma \rangle \rightarrow \langle x = 0 + 10, \sigma \rangle} \quad \frac{}{\langle x = 0 + 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle} \\
 \hline
 \langle x = y + 10, \sigma \rangle \rightarrow^* \langle \text{skip}, [10/x]\sigma \rangle
 \end{array}$$

Resources

1. Chapter Small-Step Operational Semantics in **Software Foundations - Volume 2**, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey
<https://softwarefoundations.cis.upenn.edu/plf-current/Smallstep.html>
2. Small-step SOS:
<http://fsl.cs.illinois.edu/images/6/65/CS422-Spring-2015-02c-SmallStep.pdf>